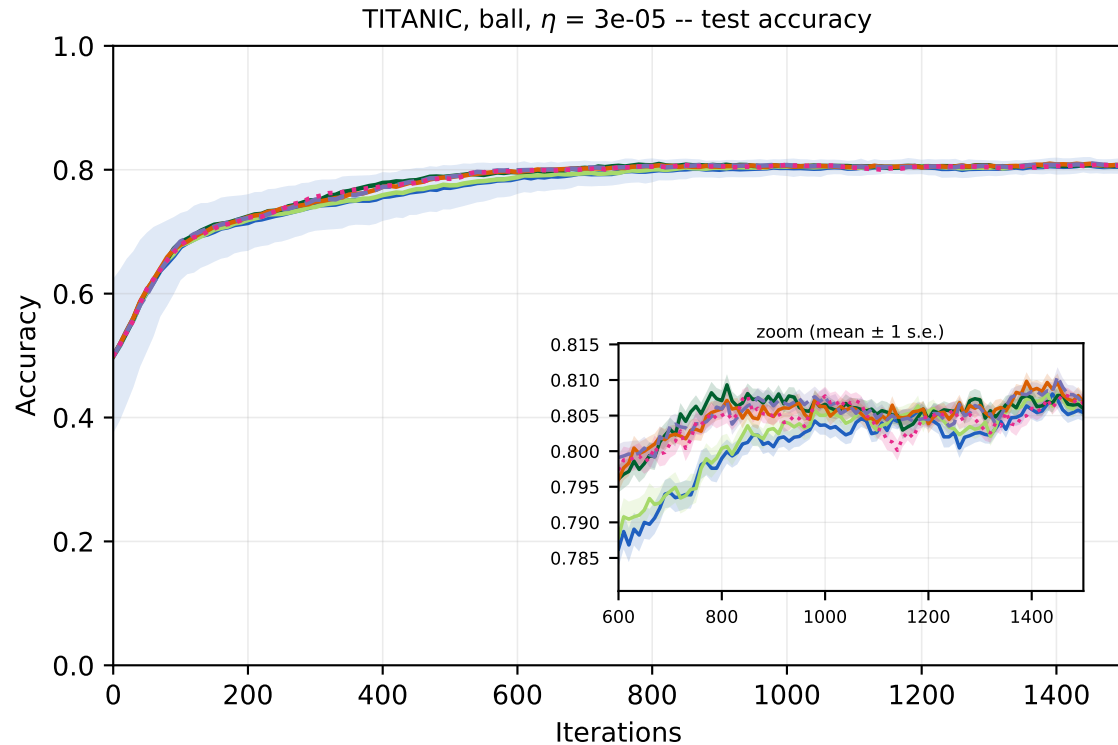
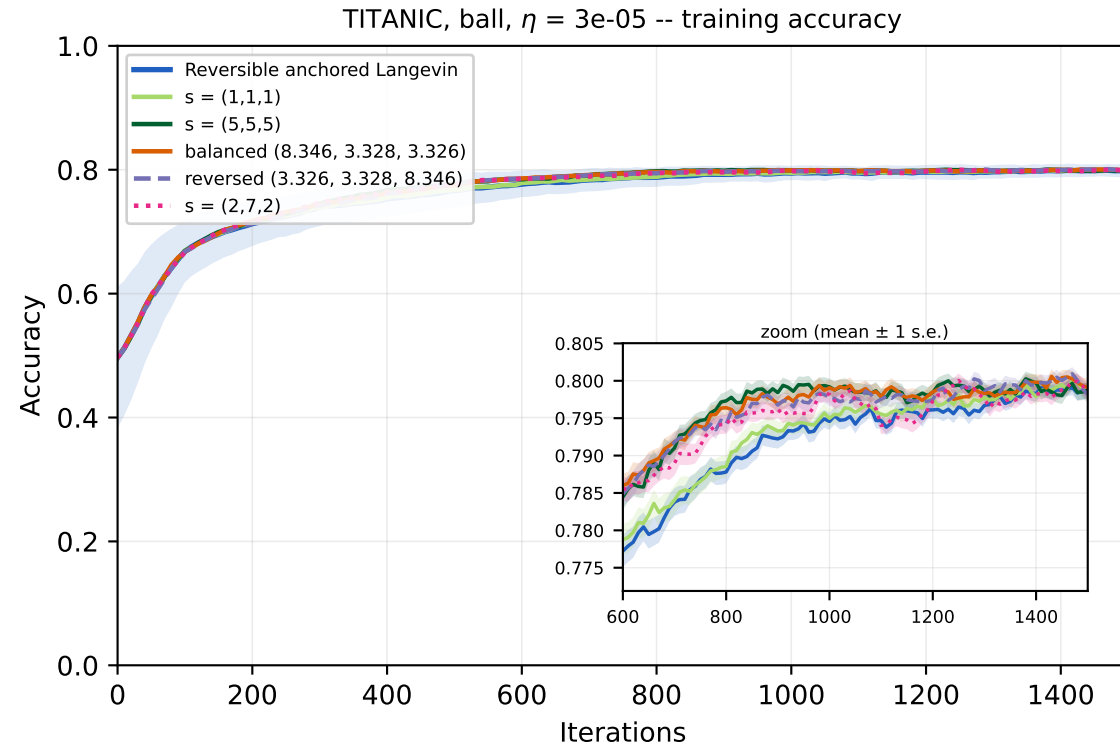


titanic_ball: per-block strengths at $\eta = 3\text{e-}05$ (mix)



TITANIC -- ball $\|w\|^2 \leq 2$ (radius $\sqrt{2}$). $d = 10$ (intercept w_0 + 9 features). $U = \sum \text{softplus} - y x \cdot w + w_0^2 / (2 \sigma^2) + \lambda \sum_{\{j \geq 1\}} |w_j|$; U_0 replaces $|w_j|$ by $\sqrt{w_j^2 + \delta^2}$; $a = \exp(U - U_0)$. EXACT gradient. $\sigma = 10$, $\lambda_{\text{lasso}} = 7.12 = 0.01 n_{\text{train}}$, $\delta = 0.0108$ (a in $[0.50, 1]$). Step $\eta = 3\text{e-}05$. Per-block strengths $s = (s_1, s_2, s_3)$ on $I_1 = \text{Age/SibSp/Parch}$, $I_2 = \text{Fare/is_female/Pclass}_2$, $I_3 = \text{Pclass}_3/\text{Embarked}_Q/\text{Embarked}_S$. Isotropic vs gradient-balanced vs reversed-balanced vs hand-picked. $R = 100$ independent replicates on one fixed stratified 80/20 split ($n_{\text{train}} = 712$, $n_{\text{test}} = 179$). Band: mean $\pm$ 1 SD (ddof = 1) for the reversible arm -- repeat-run variability at a fixed split, not a confidence or credible interval. All triples are reported in results/blocks/blocks.csv; nothing here was selected.